



Lead-cnn: lightweight enhanced dimension reduction convolutional neural network for brain tumor classification

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Abstract

In many nations, brain tumors are among the most serious medical conditions affecting both adults and children. Timely and accurate diagnosis is crucial, as the life of a patient may be shortened if brain tumors are not identified promptly. When brain tumors are diagnosed accurately and promptly, patients can receive the appropriate treatment. One of the challenging aspects of dealing with brain tumors is their diversity, making it difficult to categorize them accurately due to their varying characteristics. Deep learning (DL) models have emerged as a valuable tool in classifying brain tumors into distinct groups. However, these models face challenges related to accuracy and computational cost, necessitating advancements in brain image classification. Our primary aim is to develop a lightweight convolutional neural network (CNN) architecture that offers high efficiency, we incorporate a modified dimension reduction block with CNN to detect brain cancers with minimal computational resources while maintaining a high level of accuracy. In this research, we introduce a novel lightweight network called LEAD-CNN, designed specifically for the accurate and reliable identification of brain cancers from magnetic resonance images (MRI). Our approach begins by evaluating the effectiveness of several pre-trained DL models, including ResNet-101, VGG-19, MobileNet-V1, and DenseNet-201, which have been recommended for the classification of medical images. Subsequently, we propose a dimension reduction base CNN architecture with a lightweight design aimed at achieving a low false-negative rate in tumor image classification for patients. To assess the performance of the proposed model, we utilize a benchmark brain MRI dataset (Kaggle) comprising 7023 images, encompassing three different types of brain tumors long with normal brain images. Our experimental results are demonstrating an overall classification accuracy of 98.70%, with an average recall, F1-score, and precision of 98.60%, 98.62, and 98.65%, respectively. These findings underscore the robustness and reliability of the LEAD-CNN model. The results reveal that the LEAD-CNN model surpasses current techniques, offering a valuable tool to aid medical practitioners in the diagnosis of brain cancers.

Keywords Brain tumor · Classification · Dimension reduction · Lightweight

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1 Introduction

The brain is essential for human activities but due to its complex structure and functions make clinical study challenging. Brain tumors are a common issue in the brain arising from the uncontrolled growth of abnormal cells. These can originate within the brain itself or as secondary (metastatic) tumors, which start in other body parts like the lungs, breasts, kidneys, or skin, and then migrate to the brain through the bloodstream [1]. These highly aggressive tumors pose a severe treatment challenge and claim extensive number of deaths globally. There are over 120 types of brain tumors including glioma, pituitary tumors, and meningioma being the most common tumors with their incidence rates of 45%, 15%, and 15%, respectively [2]. Accurate detection of brain diseases is essential in today's world. While a variety of medical imaging tools exist but MRI emerges as an effective mechanism due to its radiation-free approach and non-invasive properties. Unlike CT-scans [3], MRI allows detailed examination of brain structures, empowering faster diagnoses and potentially better outcomes for patients [4]. The approach employs by utilizing deep CNN enables precise brain tumor classification from MRI scans, offering potential improvements in clinical diagnosis and treatment planning [5].

Medical imaging techniques serves as a crucial diagnostic tool in the detection of various life-threatening diseases with remarkable precision [6–12]. In the realm of brain tumors, the preferred imaging modality is magnetic resonance imaging (MRI), offering non-invasive, painless, and precise visualization. However, manual inspection of the extensive amount of MRI data for numerous patients is time-consuming and introduces potential human errors. To address this, advanced computational methods [13, 14] are being explored to efficiently analyze MRI images and optimize the patient outcomes precisely.

Brain tumor classification methods fall into ML and DL-based methods. Traditional ML-based approaches require manual feature extraction from training images, relying on expert knowledge to identify key features. This makes ML-based methods less accurate and error-prone, especially with larger databases [15]. In contrast, DL-based algorithms [16, 17], especially CNNs have gained popularity in various applications, including medical imaging. CNNs automatically extract both low- and high-level features from data, making them more effective and interesting to researchers [18–20]. Traditional CNNs for brain tumor classification are computationally intensive, leading to longer inference times and requiring a higher demand for resources. Lightweight CNNs are essential to mitigate these issues, as they provide faster processing and require fewer resources, making them more suitable for real-time medical

image analysis in resource-constrained environments [21–24].

In this study, we specifically focused on developing a lightweight model, prioritizing a minimal parameter count while achieving competitive accuracy. Consequently, we excluded transformer-based architectures and hybrid models from our comparative analysis due to their inherently large parameter counts. This exclusion was necessary to maintain a fair comparison with our proposed model, which is designed to be more parameter-efficient.

In this research, we introduce a novel dimension reduction base LEAD-CNN approach designed specifically to enhance the for multi-class brain tumor classification using MRI images. The primary aim behind developing this novel LEAD-CNN is to achieve superior performance while maintaining a rationalized architecture with fewer layers and reduced parameter calculations compared to existing state-of-the-art methods. Our methodology was precisely trained and assessed utilizing a publicly accessible MRI dataset, employing various quantitative metrics including accuracy, precision, and recall. The key contributions of our approach include:

- The study introduces a novel DL model known as LEAD-CNN, specifically designed to enhance the process of detecting brain tumors in MRI images. LEAD-CNN is custom-built to address the challenges associated with tumor identification, exploiting the power of CNN to achieve precise classification.
- In this paper, LeakyReLU activation is employed as a solution to mitigate issues related to dead neurons, rather than relying on traditional zero-centered ReLU functions. By adopting these techniques, the LEAD-CNN model effectively overcomes problems related to zero-centered issues, resulting in improved classification performance and reduced convergence time.
- Dealing with high-dimensional data can be computationally intensive. Compared to activation functions that exhibit completely flat or zero gradients for negative inputs, the proposed LEAD-CNN architecture, enhanced with modified dimension reduction blocks, substantially reduces the number of parameters and computations required in subsequent layers. This optimization leads to faster convergence during the training process.

2 Related work

Brain tumor classification is crucial given the clinical complexity. This section reviews prior work, classifying approaches into ML and DL-based methods. ML methods

require manual feature extraction, limiting accuracy due to feature quality [25]. Biswas et al. [26] tackled the challenge of efficient medical image indexing and retrieval. They proposed a novel approach employing the SPT and LNDP. They conducted experiments on two distinct medical image datasets, demonstrating noteworthy improvements over existing methods but the limitation pertains sensitivity to variations in image quality or noise. Singh et al. [27] resolve the challenge of MR brain tumor classification, employing ICA and a K-SVM methodology, achieving an impressive accuracy (98.87%). The generalizability of the approach to diverse datasets or variations in MRI image characteristics is still challenging. Gokulalakshmi et al. [28] enhanced brain tumor detection in MRI Images through a Robust Model Integrating Discrete Wavelet Transform-Based Feature Extraction and an SVM Classifier. Evaluation with simulation data yielded an impressive 94.2% accuracy, but potential limitations in generalizing to diverse datasets should be addressed.

In the study by Alnaggar et al. [29], they utilized Hyper-Layer-CNN to perform feature extraction and employed the Hyper-Heuristic-ELM for tumor classification. Their approach yielded an impressive accuracy (95.89%). Dehkordi et al. [30] conducted a comparative study between classical machine learning methods and deep learning techniques for brain tumor classification. Using datasets from BRATS 2015 and Harvard Medical School their optimized CNN model achieved an outstanding accuracy of 97.4%, surpassing the performance of other models. Pitchai et al. [31] propose a novel approach for brain tumor segmentation using DL and fuzzy K-means clustering in MR. Their methodology employs an ANN alongside the Fuzzy K-means algorithm, resulting in a notable accuracy (94%). The accuracy can be improved by using some other algorithms. Lu et al. [32] enhance abnormal brain condition detection in MRI using a modified AlexNet and ELM optimized through a CBA, achieving an impressive accuracy (96.43%). However, the limitation lies in the system current capability to classify brain images only as abnormal or healthy, with future research aiming to address multi-class classification. Murthy et al. [33] introduce a brain tumor classification model that combines AFDF-based Segmentation with an OCNN-EC. Utilizing data from the Kaggle dataset, the model achieves an accuracy (85%), highlighting its performance in brain tumor classification.

Study [34] introduced BrainMRNet, a novel CNN for brain tumor detection in MRI images, achieving a 96.05% classification accuracy by surpassing pre-trained CNNs. A novel CNN-based neural network approach for brain tumor classification in MRI images was introduced by Cinar et al. [35]. Their method achieved exceptional accuracy, detecting Glioma tumors with 99.64% accuracy

and Meningiomas tumors with (96.53%) accuracy. Waghmare et al. [36] utilized the VGG16 architecture for the classification of brain tumor MRI images, implementing preprocessing and data augmentation techniques before inputting the data into the CNN. The system demonstrated an impressive accuracy (95.71%). Mzoughi et al. [37] developed a CNN model with the purpose of distinguishing between HGG and LGG in MRI images extracted from the BraTS-2018 dataset. The system attained an impressive accuracy (96.49%). Despite significant efforts in this field, discovering an effective and applicable approach for brain MR image classification remains an ongoing challenge. One notable limitation in the studies mentioned is their exclusive focus on binary tumor classification, omitting the crucial aspect of multiclass classification to identify specific tumor types. Further investigation is essential to address this gap and enhance the classification strategy.

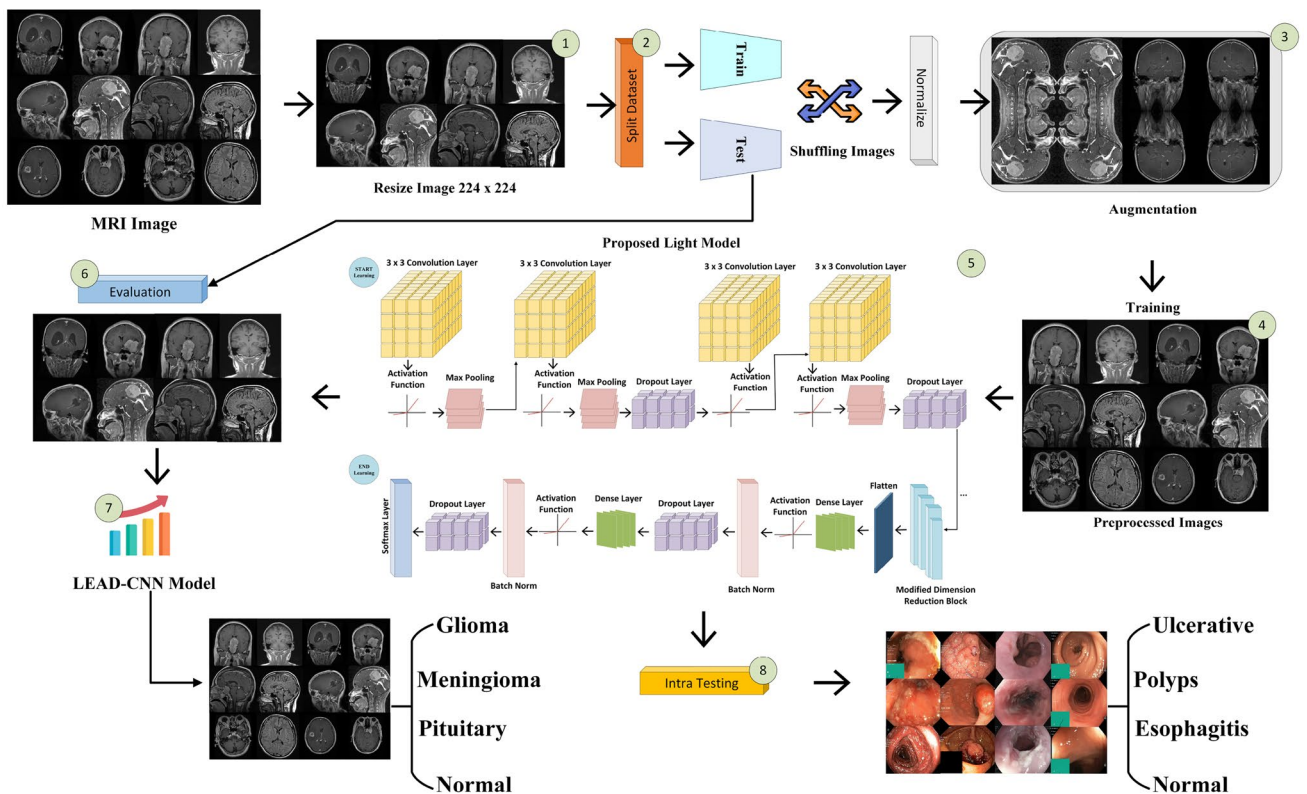
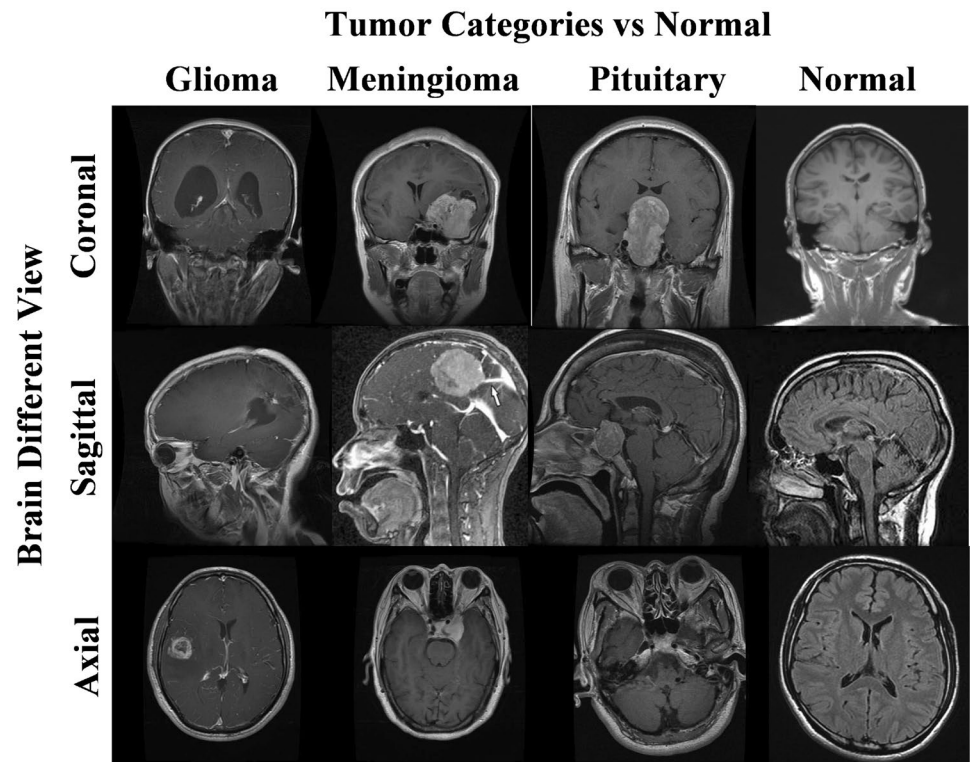
3 Materials and methods

Our proposed model primarily aims to automate the differentiation between normal MRI scans and those displaying tumor presence. The approach we propose, known as the LEAD-CNN method for classifying three tumor types in MRI images, is detailed in Fig. 1. In this section, we will elaborate on each step of our brain tumor detection methodology, which includes data collection, image pre-processing, training transfer learning models utilizing well-established TL neural networks, and the architecture design of our suggested LEAD-CNN model. The following subsections will provide a more in-depth insight into each stage of our approach.

3.1 Brain Tumor Dataset

Accurate prediction of brain tumors is importance due to its potential to enable prompt identification and treatment. Failing to diagnose brain tumors adequately can lead to detrimental consequences, including heightened tumor development and metastasis, ultimately resulting in more severe health complications and reduced treatment options. To facilitate this crucial task, we leveraged a dataset sourced from the publicly available Kaggle repository [38], comprising 7023 brain MRI images. This dataset encompasses four distinct categories of images: pituitary (1757), meningioma (1645), glioma (1621), and normal (2000). To offer a glimpse into the dataset, Fig. 1 showcases an example of brain MR images, all with a standard resolution of 512×512 pixels. The foundation for delivering proper medical

Fig. 1 Random sample images of different tumor class and normal MRI



care and enhancing patient outcomes hinges significantly on timely diagnosis and precise classification.

3.2 Proposed methodology

DL techniques have demonstrated superior performance compared to traditional ML approaches, particularly in tasks such as diagnosing potential brain tumors using MRI scan images. These DL models have shown significant improvements in outcomes. However, they still suffer from issues like over-fitting, high parameter counts, lengthy training times, and expensive practical application. Our main aim in developing the LEAD-CNN architecture was to distinguish between different tumor patients based on MRI scan images while minimizing detection time and maintaining high accuracy. The key advantage of this network lies in its lightweight and simple architecture, which boasts fewer parameters compared to other models. This approach helps mitigate over-fitting and substantially reduces computational costs. Figure 2 illustrates the LEAD-CNN model work flow with all steps, which relies on fundamental components. In the following section, we will provide detailed descriptions of each component, along with information about their respective parameters.

- **Step 1: Standardized Image** For optimal results, it is recommended to standardize the dimensions of the MR images in the dataset. The images originally come in two sizes: 512×512 and 236×236 . However, different neural network models have varying input size requirements. For example, the Xception architecture expect input images to be of size 299×299 , while the DenseNet201, and ResNet152V2 models prefer images resized to 224×224 . To ensure compatibility with all the neural network architectures used in this study, we have resized all the brain tumor images to a common dimension of 224×224 . This allows us to maintain consistency across the different models, enabling fair and accurate comparisons during our research.
- **Step 2: Dataset Distribution** In the domain of image classification, effective data partitioning plays a crucial part in the training phase of DL models. This crucial step

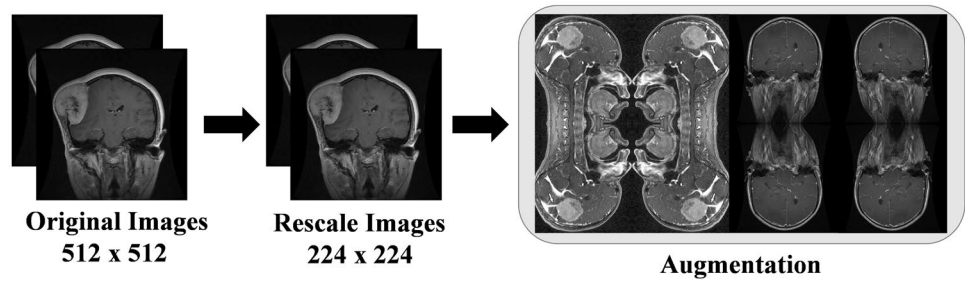
involves dividing the image dataset into three distinct sections: training, testing, and validation. Typically, a widely employed split ratio is 80–20, adhering to the Pareto principle [39]. This means that images (20%) are reserved for testing purposes, while the 80% remaining are utilized for training and validation, as it has been widely adopted in prior research. In the field of medical image analysis, particularly [14, 40]. The distribution of data for training and testing the model is summarized in Table 1, which also provides essential information about the images contained within the dataset. To enhance computational efficiency and ensure uniformity, all images have been uniformly resized to a standardized dimension of 224×224 pixels

- **Step 3: Preprocessed Image** Efficient data preparation is crucial when working with medical imaging datasets. In our study, we employed a dataset comprising 7023 images and conducted experiments with fix input image size and enhancing the performance of our LEAD-CNN model can be achieved through various pre-processing techniques. The dataset was divided into test, validation, and training, in order to ensure a robust evaluation of our model. One of the primary challenges in training DL models on medical data is the risk of overfitting, especially when dealing with limited data. To address this issue, we utilize data augmentation techniques to augment our images and gain the generalization capabilities of the LEAD-CNN model across a variety of scenarios. Specifically, we utilized the ImageDataGenerator class from the Keras library to apply data augmentation techniques to our images. By incorporating these augmentation methods into our training pipeline, we aimed to enhance our LEAD-CNN model's resilience to different input size and reduce the risk of overfitting.
- **Step 4: Augmentation** To mitigate the risk of overfitting from a limited number of images in the dataset, we employ a strategy known as image augmentation. This approach ensures that each DL model benefits from a sufficient number of input images without the need to gather additional data. Rather than collecting more data, we enhance the performance of the DL model by increase the existing dataset. To enhance our study, we have integrated two crucial augmentation techniques, one of which involves rotating the content by 90° . All images in the dataset are rotated by 90° . This augmentation not only increases the variety of input images but also helps the model generalize better by exposing it to different orientations of the same objects. Horizontal flipping: Another technique we employ is the horizontal flipping of all images. This mirrors the images along a vertical axis, further mix the dataset and aiding the model in learning features from both the original and

Table 1 Distribution of dataset into training and test sets

Class	Train	Test	Total
Glioma-tumor	1321	300	1621
Meningioma-tumor	1339	306	1645
Pituitary-tumor	1457	300	1757
Normal	1595	405	2000
Total	5712	1311	7023

Fig. 3 An overview of pre-processed and augmented MRI images



Improved Dimension Reduction Block Architecture Diagram Overview

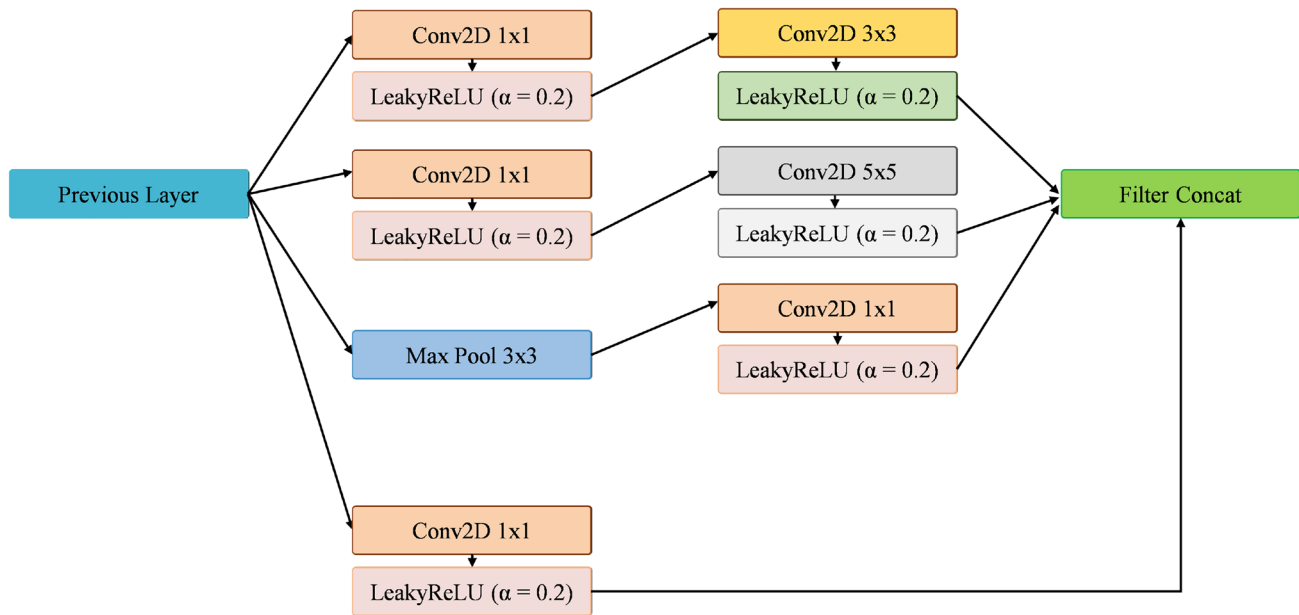


Fig. 4 An overview of Improved Dimension Reduction Block Architecture Diagram

mirrored perspectives. A visual representation of the preprocessing and augmentation steps can be found in Fig. 3, illustrating the practical implementation of these techniques to enhance the robustness and performance of our LEAD-CNN model.

- **Step 5: Proposed LEAD-CNN model** Compared to conventional ML techniques, CNN methods have consistently demonstrated superior performance, particularly in the critical task of diagnosing potential brain tumors from MRI scan images. However, DL models often challenges. Our primary objective in developing the LEAD-CNN architecture is to streamline the process of categorizing patients with diverse tumors based on their MRI images. We aim to reduce detection time without compromising accuracy. One of the key advantages of our network lies in its simplicity and reduced complexity when contrasted

with other models. In this section, we present the architecture of LEAD-CNN, which incorporates essential components for brain tumor classification. Figure illustrates the architecture overview of our proposed LEAD-CNN model. In the subsequent subsections, we will provide a comprehensive description of each component within our network, including the relevant parameters associated with them.

- **Input layer** The dimensions of an image play a crucial role in determining the input layer of our model. In our dataset, MRI images exhibit varying sizes. To ensure consistency across our network, it is essential that every image is standardized to a fixed dimension of 224 by 224 pixels.
- **Convolution layer** The convolutional layer is a crucial foundational component within CNN, where

Table 2 Proposed LEAD-CNN network layer by layer overview

Layer name	Layer type	Output shape	Parameters
Input	Input layer	$224 \times 224 \times 3$	0
Conv2d 1st	Conv2D layer	$222 \times 222 \times 32$	896
Activation function	LeakyReLU layer	$222 \times 222 \times 32$	0
Max pooling	Max Pooling 2d	$111 \times 111 \times 32$	0
Conv 2nd	Conv2D layer	$109 \times 109 \times 32$	9248
Activation function	LeakyReLU layer	$109 \times 109 \times 32$	0
Max pooling	Max Pooling 2d	$54 \times 54 \times 32$	0
Dropout (0.25)	Dropout	$54 \times 54 \times 32$	0
Conv2d 3rd	Conv2D layer	$52 \times 52 \times 64$	18,496
Activation function	LeakyReLU layer	$52 \times 52 \times 64$	0
Conv2d 4th	Conv2D layer	$50 \times 50 \times 64$	36,928
Activation function	LeakyReLU layer	$50 \times 50 \times 64$	0
Max pooling	Max Pooling 2d	$25 \times 25 \times 64$	0
Dropout (0.25)	Dropout	$25 \times 25 \times 64$	0
Conv2d 5th	Conv2D layer	$23 \times 23 \times 128$	73,856
Activation function	LeakyReLU layer	$23 \times 23 \times 128$	0
Conv2d 6th	Conv2D layer	$21 \times 21 \times 128$	147,584
Activation function	LeakyReLU layer	$21 \times 21 \times 128$	0
Max pooling	Max pooling 2d	$10 \times 10 \times 128$	0
Dropout (0.25)	Dropout	$10 \times 10 \times 128$	0
Conv2d 7th	Conv2D layer	$10 \times 10 \times 16$	2064
Conv2d 8th	Conv2D layer	$10 \times 10 \times 16$	2064
Activation function	LeakyReLU layer	$10 \times 10 \times 16$	0
Activation function	LeakyReLU layer	$10 \times 10 \times 16$	0
Max pooling	Max pooling 2d	$10 \times 10 \times 128$	0
Conv2d 9th	Conv2D layer	$10 \times 10 \times 16$	2064
Conv2d 10th	Conv2D layer	$10 \times 10 \times 16$	2320
Conv2d 11th	Conv2D layer	$10 \times 10 \times 16$	6416
Conv2d 12th	Conv2D layer	$10 \times 10 \times 16$	2064
Activation function	LeakyReLU layer	$10 \times 10 \times 16$	0
Activation function	LeakyReLU layer	$10 \times 10 \times 16$	0
Activation function	LeakyReLU layer	$10 \times 10 \times 16$	0
Activation function	LeakyReLU layer	$10 \times 10 \times 16$	0
Concatenate	Concatenate layer	$10 \times 10 \times 64$	0
Flatten	Flatten layer	6400	0
Dense	Dense layer	128	819,328
Activation function	LeakyReLU layer	128	0
Batch norm	Batch norm	128	512
Dropout (0.25)	Dropout	128	0
Dense	Dense layer	64	8256
Activation function	LeakyReLU layer	64	0
Batch norm	Batch norm	64	256
Dropout (0.5)	Dropout	64	0
Dense	Dense layer	4	260

it plays a pivotal role in shaping the model's computational process. Key parameters and hyper-parameters associated with this layer include the number of filters, and the 'kernel size.' In this con-

text, let's divide into a proposed CNN architecture comprising 12 interconnected convolutional layers, all linked to the initial input layer. In the initial two convolutional layers, a 3×3 kernel size was chosen, and each layer employed 32 filters. Moving on to the third and fourth convolutional layers, a 3×3 kernel size was retained, but these layers were equipped with 64 filters each, and so forth. These architectural choices define how the CNN processes and extracts features from the input data, ultimately contributing to the network's ability to learn and make predictions.

- *Pooling layer* A pooling layer plays a crucial role in enhancing the spatial invariance of a neural network by downscaling the feature maps. Typically positioned after a convolutional layer, this layer not only aids in reducing overfitting but also expedites the computation process. In the realm of pooling layers, there are three fundamental types: average pooling, global pooling, and maximal pooling. In our study, we have chosen to implement the maximum pooling layer, which effectively shrinks the size of the feature map and outputs the maximum value within a specified window. It's essential to note that each convolutional layer in our model is paired with a max pooling layer. As a result, our suggested architecture incorporates five max pooling layers, all employing a 2×2 window for pooling. This contributes to the downscaling and abstraction of the feature maps.
- *Dropout layer* A Dropout Layer is a widely used regularization technique in neural networks, designed to resolve overfitting problem. During training, a random subset of neurons is temporarily dropped out or set to zero. This approach promotes the network's ability to acquire a wider range of robust and diverse features, preventing neurons from co-adapting too closely. The primary advantage of incorporating a Dropout Layer is its ability to enhance a model's generalizability. By reducing the network's reliance on any specific subset of neurons, it improves the model's capacity to adapt to new data and encourage a more resilient and varied feature representation. This, in turn, reduces the risk of overfitting, a common problem in ML where a model becomes overly specialized to the training data and struggles to perform well on unseen data. Dropout encourages the network to learn a broader range of features, ultimately making it more versatile and capable of handling a variety of input scenarios.
- *Dimension reduction block* In our initial experiment, we employed the traditional ReLU activation function to construct the Naive Dimension

Reduction Block. However, as our model evolved, we encountered a common challenge (vanishing gradients) encountered in DNN due to the ReLU function. Recognizing the need for a solution, we made a strategic adjustment to our block by replacing ReLU with the LeakyReLU activation function. This alteration has proven to be a significant enhancement to our model's overall robustness and effectiveness. By adopting LeakyReLU, we have effectively mitigated the fading ReLU problem, thereby preventing the occurrence of vanishing gradients. This change has also provided us with better control over the intricacies of DNN, ultimately contributing to the improved performance and reliability of our model. Figure 4, illustrating the improved dimension reduction block architecture diagram overview to enhance the robustness and performance of our LEAD-CNN model.

- *Flatten layer* The convolutional layer and the dense layer are linked via a crucial intermediary step: flattening the input. To enhance our model's architectural structure, we introduced a dedicated flattening layer following the pooling layer, effectively extending the network's depth and versatility.
- *Fully connected layer* For the ultimate classification step, we incorporate a densely connected layer within our CNN architecture, which comes after the convolution and pooling stages. In this layer, the inputs from the preceding layers undergo a transformation, transitioning from matrices to vectors. Following this flattening process, a fully connected layer, related to a traditional neural network, takes in the transformed volume of data as its input. This fully connected layer is also referred to as a dense layer, and it plays a pivotal role in identifying features that closely correspond to specific classes within the dataset. It explore the output generated by the preceding layers, discerning patterns and relationships that are indicative of

particular class attributes. In essence, the dense layer serves as a crucial component in making predictions. One might also encounter the term fully linked layer used interchangeably with the dense layer. The activation function within this layer is responsible for producing class scores as its output, which are instrumental in the final classification. Particularly, when CNNs are employed for mapping spatial features to image labels, these dense layers become indispensable components of the architecture.

- *Activation function* The activation function is a pivotal component in augmenting a neural network's ability to grasp complex patterns by introducing non-linearity. In our approach, we leverage convolutional layers in combination with the LeakyReLU function (Eq. 1). This strategic choice addresses the problem of vanishing ReLU gradients, thereby significantly improving the performance of DNN. Unlike the standard ReLU activation function, which exhibits a flat slope for negative values, Leaky ReLU introduces a small, non-zero slope for these negative inputs. This minor alteration allows the network to better capture and learn from negative gradients, contributing to more effective and robust learning in complex tasks.

$$f(a) = \begin{cases} a, & \text{if } a \geq 0 \\ 0.01a, & \text{if } a < 0 \end{cases} \quad (1)$$

- *Step 5–8: Model Evaluation with Test Dataset* In this section, we present a series of experiments conducted on a dataset to assess the performance of both conventional TL models and our proposed LEAD-CNN model. Our experimental investigation focused on the analysis of MRI images for the identification of three tumor classes. Each pre-trained model and the LEAD-CNN model were individually trained as described above.

Algorithm 1 Introducing the LEAD-CNN Algorithm for Enhanced Brain Tumor Detection Network

Require: Magnetic Resonance Scan Image Multi Tumor (Glioma, Meningioma, Pituitary, Normal)

Ensure: Classify each category with high accuracy

- 1: **procedure** BRAINTUMORDETECTION
- 2: Load MRI Scan
- 3: Resize images from 512×512 to standard CNN size 224×224
- 4: Dataset Division (training set, testing set), and assign corresponding class label
- 5: Train set $\leftrightarrow$ Augmentation (Horizontal Flip, Vertical Flip, Rotation)
- 6: Hyperparameter setting (Callbacks, Batch Size, Epochs, Loss Function, Learning Rate, Optimizer)
- 7: LEAD-CNN (224×224 , 3)
- 8: Train Light Network with Optimal Hyperparameter setting
- 9: Store LEAD-CNN learned prediction weights into drive
- 10: Load the Saved LEAD-CNN with optimal weights
- 11: Import test MRI scan images
- 12: Light Network Model Evaluation
 1. Confusion matrix visualization
 2. Classification performance
 3. Report of Classification
 4. Cohen kappa error rate analysis

13: **end procedure**

Table 2 presents the layer types and settings employed for training the proposed LEAD-CNN model, which is designed for the analysis of both normal and tumor samples in this research. The Softmax activation function (Eq. 2) is applied as follows:

$$\text{softmax}(a_i) = \frac{e^{a_i}}{\sum_{k=1}^N e^{a_j}} \quad (2)$$

Table 2 provides an overview of the input and dataset classes. Additionally, it offers an insightful representation of the output layer, complete with parameters, which aids in comprehending the proposed LEAD-CNN model architecture.

Table 3 The hyperparameters employed in the proposed LEAD-CNN model and the pre-trained model

Parameter	Light network	Pre-trained models
Image shape	224×224	224×224
Shuffling MRI scan	During all epochs	During all epochs
Optimizer	Adam, SGD, RMSprop	Adam
Size of batches	8, 16, 32, 64, 128	64
Rate of learning	$1e-2$, $1e-3$, $1e-4$	$1e-3$
No of epochs	20, 30, 50, 100	20
Flip	Both flip (horizontal, vertical)	Both flip (horizontal, vertical)
Activation function	Final layer with softmax	Final layer with softmax

Table 4 Proposed LEAD-CNN performance across various configurations

Models	Accuracy	Precision	Recall	F1-score
CNN with ReLU	97.10	96.94	96.90	96.92
CNN with LeakyReLU	97.40	97.23	97.25	97.24
CNN with base dimension reduction block	97.48	97.30	97.40	97.34
CNN with modified dimension reduction block	98.70	98.65	98.60	98.62

Bold values means the best values

This illustration serves resource for gaining a deeper understanding of the classification process and the model's underlying structure. It plays a important role in unraveling the intricacies of the LEAD-CNN model and its capabilities in categorization.

4 Results and discussion

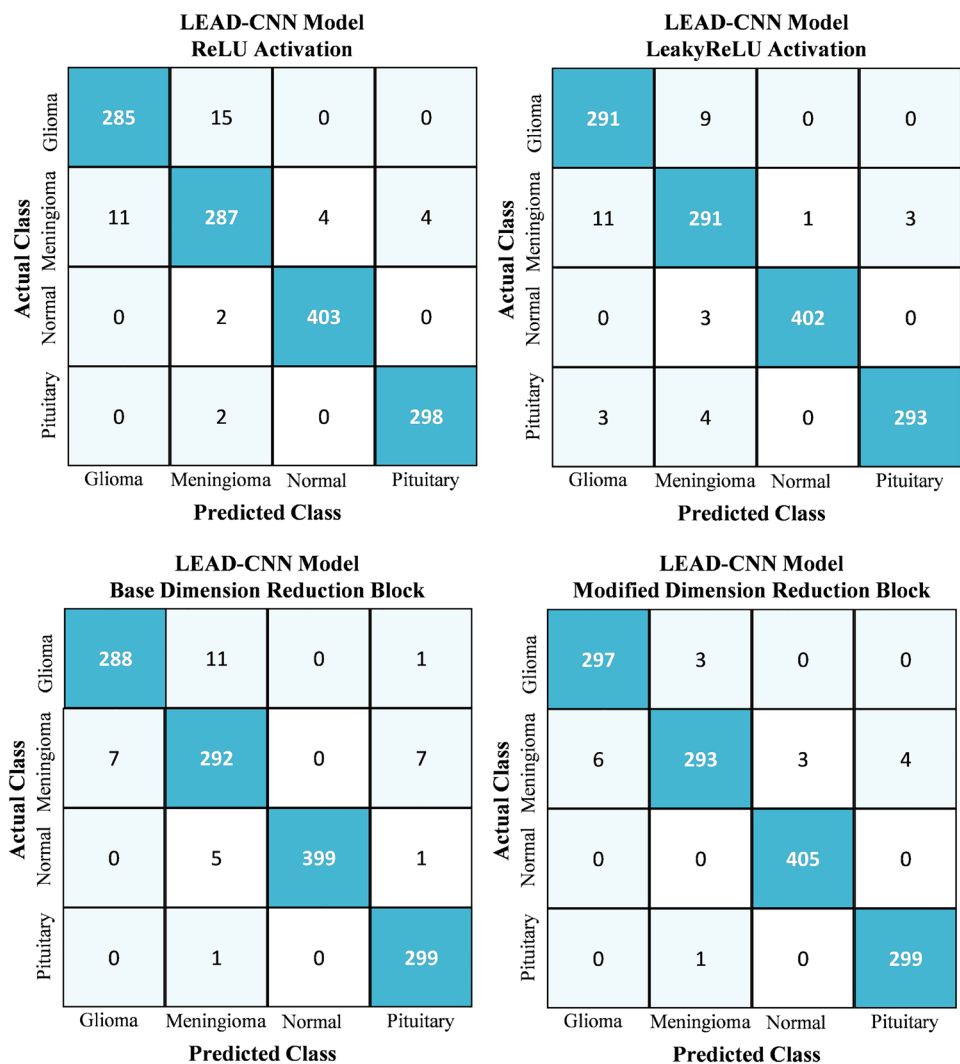
In this section, a series of experiments conducted on a dataset to assess the performance of various TL models, including the proposed LEAD-CNN model. We conducted a comprehensive experimental analysis using MRI images to classify different tumor classes. Each pre-trained model, as well as the LEAD-CNN model, underwent individual training following the previously described methodology. The implementation of this task utilized TensorFlow and Python, with the Keras package. For computational

resources, we employed a Tesla P100-PCIe graphics card with 16 GB of VRAM, an Intel Core i8 processor, 32 GB of RAM, and a 128 GB hard drive. By conducting these experiments under these hardware and software conditions, we aimed to evaluate the performance of the models and gain insights into their effectiveness for the task of tumor classification in MRI images.

4.1 Model hyperparameters

The aim of this research is to develop a streamlined and exceptionally precise model, characterized by minimal hyperparameters, to effectively classify individuals with brain tumors. To achieve this objective, it is essential to determine the optimal hyperparameter configuration. The pretrained model and the proposed LEAD-CNN model are trained using the hyperparameter settings outlined in Table 3.

Fig. 5 An overview of confusion matrix for LEAD-CNN with various configurations



4.2 Performance metrics

Evaluation metrics are essential tools for determine the effectiveness of DL categorization models once the training phase is completed [41]. To assess and compare the performance of each model on the test dataset, we employ four key performance measures: accuracy (Eq. 3), precision (Eq. 4), recall (Eq. 5), and F1-score (Eq. 6). Additionally, we consider negative-TN, false positive-FP, and false negative-FN results as part of our analysis. These metrics provide insights into the model's ability to correctly classify brain tumor images and non-tumor images. Specifically, in parentheses, we denote the count of brain tumor images incorrectly classified as non-tumor (FP), the count of non-tumor images misclassified as containing tumors (FN), and the count of brain tumor images incorrectly identified as non-tumor (FN).

$$Accuracy := \frac{TP + TN}{TP + FP + TN + FN} \quad (3)$$

$$Precision := \frac{TP}{TP + FP} \quad (4)$$

$$Recall := \frac{TP}{TP + FN} \quad (5)$$

$$F1Score := 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

Fig. 6 An overview of the Kappa coefficients and error levels in relation to various configurations of LEAD-CNN

4.3 Classification results

To assess the classification performance of the LEAD-CNN model and showcase its effectiveness, this study conducted a comprehensive set of multiclassification experiments using a brain tumor dataset. In this section, we present the experiments carried out to assess the performance of the LEAD-CNN model on brain tumor dataset. To begin with, we applied the suggested LEAD-CNN model to a dataset that featured numerous dense layers, dropout layers, and a pooling layer without any dimension reduction. The test accuracies obtained both with and without the dimension reduction block are summarized in Table 4. Our findings indicate that by replacing the base ReLU activation function with LeakyReLU, we were able to significantly enhance the test accuracies of the LEAD-CNN model. Specifically, we observed improvements of 97.40% and 98.70%, respectively, with and without the dimension reduction block. This suggests that incorporating a modified dimension reduction block can have a positive impact on the performance of the proposed LEAD-CNN models. It's worth highlighting that the inclusion of this dimension reduction block in the final iterations of the model was considered crucial in the development of the LEAD-CNN model.

The table presents the results of the suggested LEAD-CNNs with different activation functions applied to both the base and modified dimension reduction blocks. When comparing these models, it becomes evident that the CNN employing the ReLU activation function achieves an impressive 97.10% accuracy, along with a 96.94% precision, a balanced 96.90% recall rate, and an 96.92% F1-score. In contrast, the CNN utilizing the LeakyReLU activation function

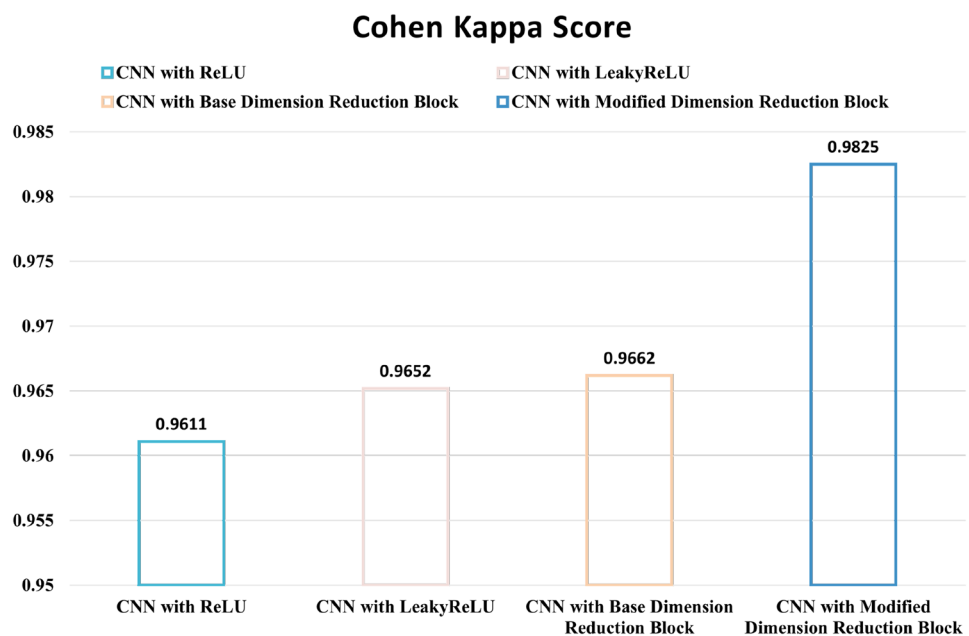


Table 5 Comparison of LEAD-CNN performance with six pre-trained models

Models	Accuracy	Precision	Recall	F1-score
DenseNet-201 (base)	96.10	95.84	95.79	95.81
ResNet-101 (base)	97.17	97.11	96.93	96.98
MobileNet-V1 (base)	94.73	94.47	94.30	94.37
Xception (base)	92.75	92.29	92.20	92.20
EfficientNet-B4 (base)	94.50	94.10	94.10	94.08
VGG-19 (base)	96.94	96.81	96.68	96.72
Proposed model	98.70	98.65	98.60	98.62

Bold values means the best values

outperforms the others, boasting an outstanding 97.40% accuracy in its predictions. When focusing on the task of brain tumor prediction, it's noteworthy that the CNN with the Base Dimension Reduction Block (second run) demonstrates superior performance, achieving the highest accuracy. This model attains an impressive 98.70% accuracy rate in predicting brain tumors. Furthermore, the LEAD-CNN employing the modified dimension reduction block emerges as the overall best-performing variant, surpassing all other combinations. It excels with a remarkable 98.70% accuracy rate in brain tumor prediction, solidifying its position as the top-performing CNN configuration.

The detailed class-wise results of the suggested LEAD-CNN model have been presented in Fig. 5 confusion matrix. By examining this matrix, we can gain insights into how well our model performed for specific classes, understanding correctly categorized images and incorrectly classified images. According to the confusion matrix, it appears that the LEAD-CNN with ReLU model had a total of 38 misclassifications on the test dataset. However, it is important to note that out of the 1349 tests conducted, our model accurately classified 1273 of them. This demonstrates a high level of accuracy in our model's predictions. Based on the evaluation metrics obtained from the proposed model's performance, it is evident that the CNN + modified dimension reduction approach outperforms other models across the board. The LEAD-CNN model consistently exhibits a 1–2% average improvement in all evaluation metrics. The LEAD-CNN model not only showcases strong performance in terms of correctly classifying images but also outperforms other models when considering various evaluation metrics. This underscores the effectiveness and robustness of the proposed approach in image classification tasks.

The results presented in Fig. 6 offer insights into the performance of various LEAD-CNN configuration models on the test set, comparing their Kappa coefficients and error levels. The analysis of these models revealed Kappa coefficients of 0.9611, 0.9652, 0.9662, and an impressive 0.9825, respectively. These models encompassed CNN +

ReLU, CNN + LeakyReLU, and CNN + base dimension configurations. Notably, the LEAD-CNN model, which was introduced, exhibited a significant enhancement in the Kappa value, reaching a coefficient of 0.9825. Moreover, the graph illustrates a clear inverse relationship between the Kappa coefficient and the error value, indicating that as the Kappa coefficient increases, the accuracy of predictions by the LEAD-CNN model improves. The LEAD-CNN model, as suggested, demonstrated a substantial boost in the Kappa coefficient, underscoring its reliability and robustness in making predictions. These findings highlight the substantial performance enhancement achieved by the proposed LEAD-CNN model, which incorporates a modified dimension block to augment the CNN model's capabilities. In terms of listed various performance evaluation, the proposed LEAD-CNN model consistently outperformed the evaluated CNN configurations. This improvement is evident in the observed gains across these performance metrics. The strength of the LEAD-CNN model lies in its ability to effectively exploit the advantages of different features and produce more accurate predictions. This is achieved through the incorporation of LeakyReLU in dimension reduction blocks, which mitigates the issue of dead neurons and enables the network to capture intricate patterns more effectively. The LEAD-CNN model stands out in its overall performance due to its skillful integration of complementary features, effectively addressing the constraints associated with individual models. These findings underscore the potential of the proposed LEAD-CNN model as a promising approach for diagnosing brain tumor illnesses.

4.4 Performance comparison with pre-trained models

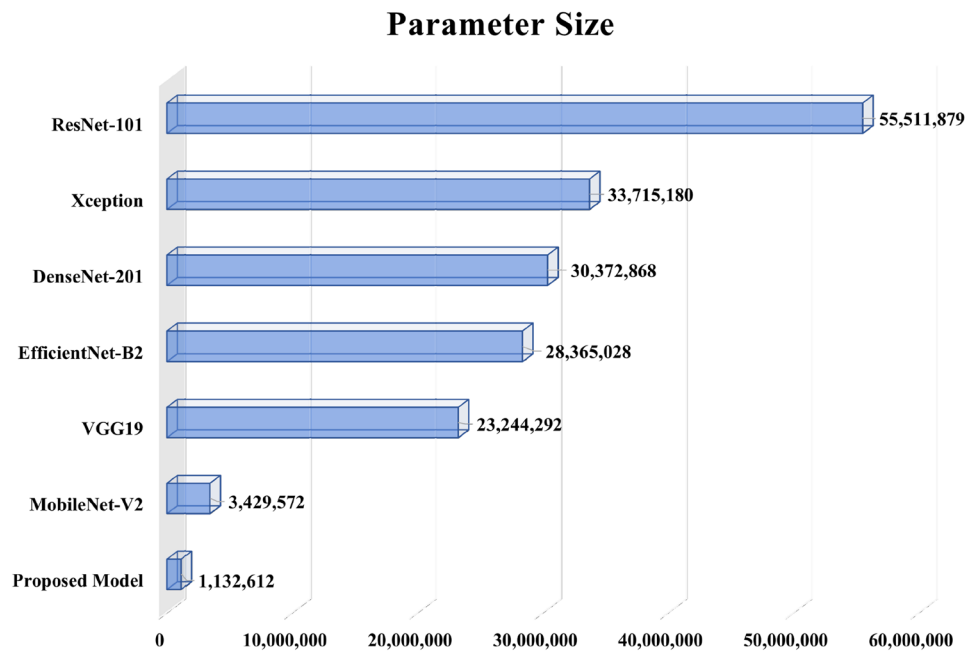
Research on brain MRI imaging for effectively predicting tumor diseases has gained significant attention, particularly with the availability of several publicly accessible brain tumor datasets. Various DL techniques have been leveraged to develop robust diagnostic models. Among

Table 6 Comparative performance analysis of proposed LEAD-CNN with prior studies

Reference	Method	Accuracy
Ehab et al. [42]	R-CNN and ANFIS	83.25%
Sharif et al. [43]	Feature Selection with DNN	92.00%
Noreen et al. [44]	Xception with KNN Ensemble	94.34%
Khan et al. [14]	Hybrid-Net Model	95.00%
Shahin et al. [45]	Modular fully CNN	96.30%
Proposed model	LEAD-CNN	98.70%

Bold values means the best values

Fig. 7 Comparing the model size of LEAD-CNN with six pre-trained models



these techniques, CNN-based models have been widely adopted, with TL being a prominent approach. However, many previous studies suffered from imbalanced data distribution, and in some cases, insufficient data hindered the evaluation of procedures. In this comprehensive study, we conducted a thorough assessment of the six most successful CNN algorithms to identify the optimal model for predicting tumor in MRI images. To achieve this, an extensive collection of tests was executed on a dataset comprising brain MRI images. These tests involved the application of six independently fine-tuned versions of pre-trained CNN models, which were originally trained on the ImageNet dataset. Our research endeavors to address the challenges associated with brain tumor prediction using MRI imaging by rigorously evaluating state-of-the-art CNN models on a well-curated dataset, ultimately aiming to enhance the prediction and reliability of tumor predictions in this critical medical context.

Table 5 displays a comprehensive evaluation of the classification results achieved by listed pre-trained model, each evaluated against same criteria. In order to mitigate the challenge of data imbalance, we ensured an equal quantity of data for each group. This table clearly demonstrates that all pre-trained models have excelled in classifying three tumor diseases and normal images. ResNet-101 stands out with remarkable precision, recall, and F1-score values of 97.11%, 96.93%, and 96.98%, respectively. This exceptional performance translates into an impressive test accuracy of 97.17%. VGG-19 follows closely behind, achieving a noteworthy test accuracy of 96.94%, along with precision, recall, and F1-score scores of 96.72%, 96.68%, and 96.71%, respectively. DenseNet201, despite a slightly lower test accuracy of 96.10%, demonstrates outstanding precision, specificity, recall, and F1-score, all surpassing the 95% (95.84%, 95.79%, 95.81%). Meanwhile, MobileNetV1, while

Table 7 Comparing the training and testing times of the LEAD-CNN model with six pre-trained models

Model	Train time	Test time	Flops (M)	Model size KB
DenseNet-201 (base)	358.33	6.70	5.52e+05 M	77,877
ResNet-101 (base)	334.29	4.72	9.72e+05 M	172,366
MobileNet-V1 (base)	314.99	3.74	7.33e+04 M	15,223
Xception (base)	341.87	6.24	5.85e+05 M	86,596
EfficientNet-B4 (base)	326.13	4.41	1.97e+05	74,396
VGG-19 (base)	336.30	3.16	2.5e+06 M	79,486
Proposed model	124.76	2.47	5.16e+04 M	13,455

Bold values means the best values

achieving a test accuracy of 94.73%, maintains strong F1-scores (94.37%, 94.47%) and precision, specificity, and recall metrics. For the specific task of brain tumor prediction, the EfficientNet-B4 Network emerges as the top performer, boasting an accuracy of 94.50%, coupled with equally impressive recall, F1-score, and precision metrics, all standing at 94.10%. On the other hand, the Xception model demonstrates the lowest performance among the considered models, with an F1-score (92.20%), recall (92.20%), precision (92.29%), and test accuracy (92.75)%.

4.5 Comprehensive review and analysis in comparison to existing studies

Table 6 provides a improved performance comparison of our proposed LEAD-CNN by conducting a comprehensive review and analysis in comparison to existing studies in the brain tumor field., likely related to DL, along with their corresponding accuracy scores. These accuracy scores represent the performance of each method in accurately accomplishing the brain tumor prediction. R-CNN and ANFIS (83.25%): this method combines R-CNN and ANFIS to achieve an accuracy of 83.25%. Feature Selection with DNN (92.00%): in this approach, feature selection techniques are applied in combination with a DNN to reach an accuracy of 92.00%. Xception with KNN Ensemble (94.34%): the Xception model, a deep learning architecture, is used along with a KNN ensemble approach to achieve an accuracy of 94.34%. Hybrid-Net Model (95.00%): this method employs a hybrid neural network model, likely combining different types of neural layers, to attain a high accuracy of 95.00%. Modular fully CNN (96.30%): a fully CNN is organized in a modular fashion to accomplish the task with an accuracy of 96.30%. while our proposed LEAD-CNN (98.70%): the LEAD-CNN model, performs exceptionally well with an impressive accuracy of 98.70%.

4.6 Complexity analysis

In Fig. 7, we can observe that the LEAD-CNN, trained using our recommended technique, outperforms all pretrained CNN models, despite having a relatively small parameter

count of precisely 1,132,612. This remarkable performance can be attributed to its shallower architecture and overall parameter efficiency. As we search deeper into neural network architectures, there is a notable exponential increase in the total number of parameters, which automatically leads to higher storage requirements. However, our success lies in optimizing the network's hyperparameters, ultimately determining the model's exceptional overall performance. The LEAD-CNN model shows an notable accuracy (97.17%), all while maintaining a relatively lightweight architecture of just 55,511,879 million parameters. This feat not only demonstrates its superior performance but also highlights its remarkable parameter efficiency when compared to even the largest model in the field, ResNet-101. This unique combination of accuracy and parameter size makes the LEAD-CNN an ideal choice, particularly for scenarios where computing resources are limited, such as remote locations with limited access to healthcare facilities. In scenarios where brain tumor testing is required in isolated or resource-constrained settings, the ideal model should strike a balance between computational efficiency, lightweight architecture, and high accuracy in tumor prediction. Unlike deep and resource-intensive models, our suggested approach offers a compelling advantage with its reduced parameter count and streamlined architecture, without compromising performance. This makes it exceptionally well-suited for deployment on edge devices and within the IoT ecosystem, significantly lowering computational costs. Furthermore, the suggested CNN model is particularly well-suited for widespread population screening. Its short training and testing times make it an efficient choice for diagnosing brain tumor cases, especially in regions with limited healthcare resources. This not only improves accessibility to critical medical services but also

Table 9 Performance of LEAD-CNN with varied learning rate configurations

Learning rate	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Cohen Kappa (%)
1e-2	96.79	96.56	96.54	96.54	95.70
1e-3	98.70	98.65	98.60	98.62	97.32

Bold values means the best values

Table 8 Performance of LEAD-CNN with varied batch size configurations

Batch size	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Cohen Kappa (%)
8	97.33	97.25	97.15	97.11	96.41
16	97.63	97.54	97.45	97.46	96.82
32	98.24	98.12	98.10	98.10	97.64
64	98.70	98.65	98.60	98.62	97.32

Bold values means the best values

Table 10 Performance of LEAD-CNN with varied epochs configurations

Epochs	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Cohen Kappa (%)
20	92.37	92.58	91.95	91.68	89.76
30	95.72	95.70	95.42	95.29	94.26
50	98.70	98.65	98.60	98.62	97.32
100	98.16	98.07	98.02	98.03	97.54

Bold values means the best values

ensures rapid and accurate diagnosis, ultimately enhancing healthcare outcomes for underserved populations.

Table 7 presents the computational costs for both training and testing times, measured in seconds. The table clearly indicates that the proposed LEAD-CNN model boasts a training time of approximately 124 s. In contrast, the other DL models encompass training periods spanning from 358 to 314 s, making them comparatively time-consuming. When considering the testing phase, the proposed LEAD-CNN boasts the shortest computation time among all models, clocking in at just 2.47 s for test data. This highlights a notable efficiency advantage. Examining the table further reveals that deep models, with their increased number of layers and trainable parameters, tend to require longer training periods. In contrast, the suggested LEAD-CNN features fewer parameters and depth, resulting in more cost-effective computational expenses. Furthermore, table provides a comparison based on their computational complexity (measured in FLOPs), parameter count, and model size. Among the models listed, the proposed model stands out for its significantly lower FLOPs (51,600,000), and compact model size (13,455 KB), making it an efficient choice compared to the

others. While models like ResNet-101 and VGG-19 exhibit higher computational demands and larger parameter counts (Fig. 6), potentially offering higher accuracy in certain tasks, the proposed model balances performance with resource efficiency, making it suitable for scenarios where computational resources are limited or where rapid deployment and inference speed are crucial. The efficiency of the proposed model could be attributed to advancements in architectural design, optimization techniques, or both, showcasing the ongoing efforts in the deep learning community to develop models that deliver competitive performance while being resource-conscious.

The lightweight nature of our LEAD-CNN proposed model is a main contribution, specifically designed to facilitate deployment on smart devices, which is crucial for real-time, on-site diagnostic support in clinical settings. By minimizing the model parameter count, LEAD-CNN (OUR) model can be efficiently implemented within the limited computational resources typically available in smart healthcare devices. This design choice not only ensures quick response times essential for clinical decision-making but also enhances the accessibility of diagnostic tools across

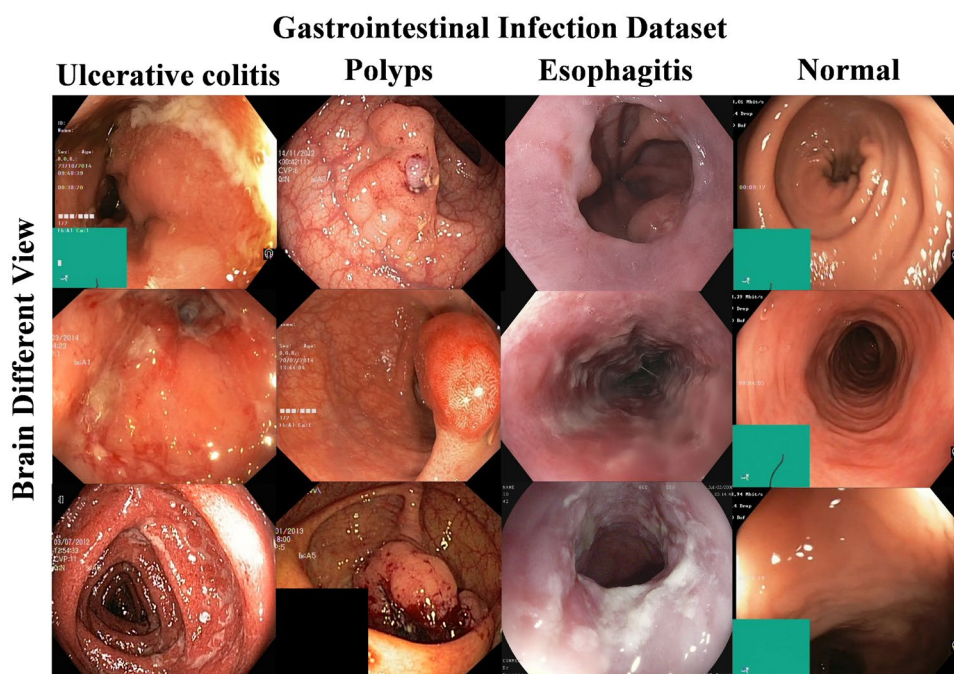
Fig. 8 An overview of gastrointestinal infections and normal samples

Table 11 Distribution of dataset into GI class

Class	Train	Test	Total
Ulcerative colitis	1300	200	1500
polyps	1300	200	1500
Esophagitis	1300	200	1500
Normal	1300	200	1500
Total	5200	800	6000

various healthcare environments. With our LEAD-CNN model, practitioners can leverage immediate, data-driven insights directly at the point of care, thereby improving the accuracy and speed of patient management and treatment. This approach focus on a parameter-efficient architecture makes LEAD-CNN suitable for scenarios where rapid and reliable diagnostic support is critical.

4.7 Unveiling insights: a study on performance ablation

Ablation studies carefully assess each hyperparameter's effect on a model's performance, they are essential. These investigations help us determine which hyperparameters are most important, adjust model configurations, and learn more about the underlying mechanisms by isolating each one and holding the others constant. In the end, this targeted strategy—where we analyze each hyperparameter separately—improves the effectiveness and performance of the model.

4.7.1 Performance analysis with batch size various

This experiment aimed to explore the impact of varying batch sizes on the enhancement of test accuracy. The outcomes are presented in Table 8, illustrating the performance of LEAD-CNN on a test dataset following training with four distinct batch sizes. The table reveals that test performance was consistently higher and exhibited less variability when a batch size of 64 was employed. Consequently, the batch size of 64 was chosen as the optimal configuration for this specific task.

Table 12 Proposed LEAD-CNN performance across various configurations

Models	Accuracy	Precision	Recall	F1-score
CNN with LeakyReLU	94	94.01	94	93.99
CNN with modified dimension reduction block	95.87	95.91	95.87	95.86

Bold values means the best values

Table 13 Comparison of LEAD-CNN performance with five pre-trained models

Models	Accuracy	Precision	Recall	F1-score
DenseNet-201 (base)	92.12	92.31	92.12	92.02
ResNet-101 (base)	85.12	86.37	85.12	84.34
MobileNet-V1 (base)	90.37	90.47	90.37	90.26
Xception (base)	92.87	92.89	92.87	92.82
VGG-19 (base)	94.87	94.84	94.87	94.84
Proposed model	95.87	95.91	95.87	95.86

Bold values means the best values

4.7.2 Performance analysis with learning rate various

This experiment aims to assess the influence of different learning rates on the performance of LEAD-CNN. The results are presented in Table 9, which displays the LEAD-CNN's performance on a test set after undergoing training with two distinct learning rates, namely $1e-2$ and $1e-3$. The table clearly illustrates that the use of a learning rate of $1e-3$ consistently yields superior test results, both in terms of performance and stability. As a result, for this study, a learning rate of $1e-3$ was chosen as the optimal parameter value.

4.7.3 Performance analysis with different epochs

This experiment search into the impact of varying the number of training epochs on the performance of the LEAD-CNN model. The results are displays in Table 10, illustrating the model's performance across four different epoch settings: 20, 30, 50, and 100. It is noteworthy that the test performance notably improved and demonstrated greater consistency when trained for 50 epochs. Consequently, epoch 50 was chosen as the optimal training duration for this research.

4.8 Additional experiments

To strengthen the validation of the proposed LEAD-CNN model, we have undertaken intra-dataset testing across different datasets, employing the gastrointestinal infection dataset for additional scrutiny. This extension of our evaluation involves replicating the experimental setup and hyperparameters employed in our previous analysis of the brain tumor dataset. This meticulous approach allows us to assess the generalizability and robustness of the LEAD-CNN model beyond its initial context. By maintaining consistency in experimental conditions and hyperparameter settings, we ensure a fair comparison between datasets, facilitating a more comprehensive understanding of the model's performance across diverse medical domains. So,

Fig. 9 An overview of confusion matrix for LEAD-CNN with various configurations

		LEAD-CNN Model LeakyReLU Activation						LEAD-CNN Model Modified Dimension Reduction Block			
Actual Class	Normal	200	0	0	0	Actual Class	Normal	200	0	0	0
	Ulcerative	0	179	21	0		Ulcerative	0	188	11	1
	Polyps	0	27	173	0		Polyps	0	21	179	0
	Esophagitis	0	0	0	200		Esophagitis	0	0	0	200
		Predicted Class						Predicted Class			
		Normal	Ulcerative	Polyps	Esophagitis			Normal	Ulcerative	Polyps	Esophagitis

in this section, this intra-dataset testing, aim to demonstrate not only the effectiveness of the LEAD-CNN model in a MRI tumor specific context but also its potential utility in addressing varied medical challenges. Through rigorous validation across different dataset, we seek to provide compelling evidence of the model's efficacy and versatility, thereby enhancing confidence in its applicability within the broader medical community.

4.8.1 Dataset details

In this study, we evaluated the performance of our proposed LEAD-CNN using a publicly available dataset of gastrointestinal-GI endoscopic images. Initially, two GI image datasets [46, 47], curated by [48], were combined to form a comprehensive GI endoscopic image dataset. We employed this dataset for training and testing the LEAD-CNN model for detecting GI disorders [3]. The dataset comprised a total of six thousand images, evenly distributed into four groups of 1500 images each to represent different infection categories. Figure 8 illustrates samples from each infection category, while Table 11 outlines the distribution of the dataset among these classes. we followed the dataset split previously applied by the authors and publicly available. This split utilized 200 samples for each class to validate the model's performance.

4.8.2 Classification performance evaluation of LEAD-CNN on diverse dataset

Integrating LeakyReLU activation with the dimension reduction block provides two significant benefits. First, LeakyReLU helps alleviate the vanishing gradient problem by allowing small negative gradients, enhancing the flow

of gradients during training and potentially leading to faster convergence. Second, coupling LeakyReLU with the dimension reduction block can improve the model's ability to capture and retain essential features while reducing computational complexity, resulting in more efficient and accurate CNN models. Table 12 presents performance metrics of proposed LEAD-CNN model configurations: one with LeakyReLU activation and another with a LeakyRelu along modified dimension reduction block. The featuring the modified dimension reduction block, exhibits higher accuracy, precision, recall, and F1-score compared to the only LeakyReLU along proposed CNN configuration. Specifically, it achieves an accuracy of 95.87%, precision of 95.91%, recall of 95.87%, and F1-score of 95.86%.

Table 13 provide a detailed performance evaluation of LEAD-CNN alongside five prevalent pre-trained models. Each model's accuracy, precision, recall, and F1-score are meticulously compared. Notably, the proposed LEAD-CNN model achieves remarkable results across all metrics, surpassing even the top-performing base models like VGG-19 and DenseNet-201. This suggests that LEAD-CNN holds promise for tasks requiring high classification accuracy, such as image recognition or medical diagnosis. Moreover, in addition to its superior performance, LEAD-CNN likely offers benefits in terms of computational efficiency and model size compared to heavier pre-trained models like VGG-19 and DenseNet-201, thus making it more suitable for deployment in resource-constrained environments or scenarios necessitating real-time inference. The Fig. 9 illustrates the confusion matrix visualization of the proposed model under two distinct configurations: one employing CNN with LeakyReLU, and the other integrating LeakyReLU within a modified dimension reduction block to streamline model complexity. By providing a visual representation of the model's performance across different classes, confusion

matrix visualization aids in understanding classification errors and strengths, facilitating targeted improvements in model optimization.

5 Strength, limitation and future work

Our innovative approach to accurately distinguish between normal and tumor-infected MRI images represents a significant contribution to the field of brain tumor detection. In large-scale screening scenarios where resources are limited, the manual identification of tumor-affected brain regions is time-taking and permitting to human error which resulting in inaccurate reporting. To address this challenge, we utilize the proposed LEAD-CNNs to automate the classification of tumor MRI, offering a more reliable and precise means of diagnosing brain tumors. Our research findings strongly support the effectiveness of the LEAD-CNN model in tumor detection, with outstanding performance metrics: a classification accuracy (98.70%), a recall rate (98.60%), precision (98.65%), and an F1 score (98.62%). These results underscore the proficiency of LEAD-CNN in accurately identifying brain tumors, positioning it as a promising technique for precise and timely tumor diagnosis.

To further validate the robustness of our proposed model, we employed an intra-testing strategy. Utilizing the widely recognized gastroinfection dataset, we ensured our model's consistency by achieving an impressive 95.87% accuracy rate. Testing a model on different intra datasets is crucial for several reasons:

- *Generalizability* Assessing the model's performance on diverse datasets ensures its ability to generalize across various scenarios. It demonstrates the model's capability to handle different data distributions and characteristics beyond the original training data.
- *Evaluation of stability* Testing on multiple datasets allows us to gauge the stability of the model's performance. Consistency across different datasets indicates a reliable and robust model, less prone to overfitting or biases specific to the training data.

Evaluation of stability: By conducting intra-testing using the gastroinfection dataset, we not only reaffirmed the effectiveness of our model but also showcased its adaptability and reliability in handling different data domains.

Despite the promising outcomes of our study, it is essential to acknowledge its limitations. We exclusively relied on a publicly available brain tumor dataset, which carries inherent limitations and potential biases owing to its scale and diversity constraints. To enhance the

generalizability of the LEAD-CNN model, more validation on larger and diverse datasets is imperative. Such an approach would provide deeper insights into the model performance across various contexts and demographic groups. Additionally, while we evaluated LEAD-CNN performance using widely used pre-trained models for tumor identification, conducting comparative research with other existing techniques would offer a more comprehensive assessment of its efficacy.

In our current work with the LEAD-CNN model, we have exclusively utilized public datasets that comply with all ethical clinical standards to safeguard patient privacy and prevent data loss. These datasets have been carefully curated to ensure that all personal health information is de-identified, aligning with legal and ethical guidelines. This approach not only reinforces our commitment to ethical AI development but also allows us to demonstrate the effectiveness of our model without compromising patient confidentiality. Through rigorous adherence to these standards, we ensure that our research remains responsible and respectful of patient rights in future work, providing a safe framework for AI integration in clinical environments. Moreover, In the future, we are committed to conducting extensive investigations into the model performance on other benchmark datasets and in real clinical settings. This effort aims to validate its reliability and practical utility, further solidifying its potential as a dependable tool for accurate and timely tumor identification.

6 Conclusion

Detecting brain tumors is crucial due to their potentially contagious and fatal nature. Recent advancements in CNN models have sparked interest in medical imaging, and our research aims to further this field with a novel approach called LEAD-CNN for MRI-based tumor diagnosis. Our CNN method employs a modified dimension reduction block that effectively categorizes images with remarkable accuracy, eliminating the need for manual feature extraction. The LEAD-CNN model is a novel framework that operates fully automatically. Our findings demonstrate that the LEAD-CNN model's robustness and generalization capabilities are significantly improved through careful selection of hyperparameters. This innovative model successfully distinguishes between healthy and infected brain images, achieving an impressive 98.70% accuracy, 98.60% recall, 98.65% precision, and 98.62% F1 score. Furthermore, our findings reveal that the LEAD-CNN model outperforms pre-trained models across various evaluation metrics, making it a superior choice for brain tumor image classification and disease screening. In real-time applications, we expect that the LEAD-CNN model will expedite and enhance patient

screening for tumor infections. Future research endeavors should focus on enhancing generalizability by expanding the dataset, exploring cutting-edge techniques such as novel block, and conducting field validation experiments to assess the model's effectiveness. To enhance the validation of our proposed LEAD-CNN model, we employed an intra-testing strategy. Utilizing the renowned gastroinfection dataset, our model consistently achieved 95.87% accuracy. Testing our model on diverse intra datasets is crucial for comprehensive validation. It ensures the robustness and generalizability of the model across various data domains, thereby enhancing its reliability in real-world applications.

Author Contributions Saif Ur Rehman Khan: conceptualization, data curation, methodology, software, validation, writing original draft. Sohaib Asif: conceptualization, funding acquisition, supervision. Omair Bilal and Hafeez Ur Rehman: supervision, validation, writing review and editing.

Data Availability Statement Data are available from the authors upon reasonable request and with permission of Author.

Declarations

Conflict of interest There is no conflict of interest.

Ethics approval Approved.

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